

Analysis of the OH Coverage on Low-Coordinated Pt Sites at Low Potentials

Rubén Rizo,* Lorena Chico-Mesa, Rosa M. Arán-Ais, Víctor Climent,* Enrique Herrero,* and Juan M. Feliu



Cite This: *ACS Electrochem.* 2025, 1, 351–359



Read Online

ACCESS |



Metrics & More



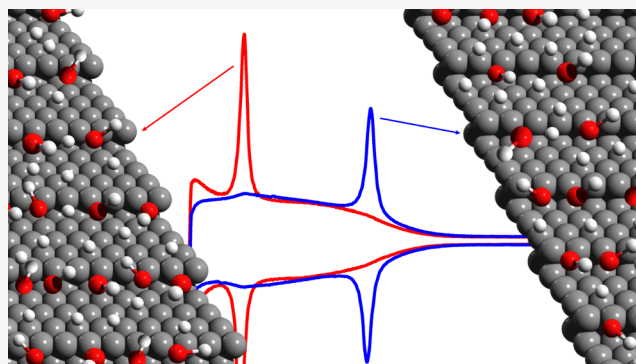
Article Recommendations



Supporting Information

ABSTRACT: In a previous study, the presence of OH species adsorbed on low-coordinated Pt sites (or steps) at low potentials (the so-called hydrogen adsorption region) using spectroscopy and electrochemical experiments was demonstrated. This investigation was specifically conducted on a Pt(311) surface in an electrolyte with pH = 1. In this work, the study is extended to other surfaces featuring (111) terraces but with varying step geometry and step density. Additionally, the influence of the pH on these systems was explored. Our findings reveal that the OH coverage on Pt steps at low potentials, measured just after the peak for the step, is independent of pH but sensitive to surface structure. Specifically, we have determined that OH coverage is approximately 0.5 on surfaces with (110) steps and 0.33 on surfaces with (100) steps. Furthermore, the potential of zero total charge aligns with the local potential of zero charge for the steps since it is predominantly influenced by adsorbed species on these low-coordinated sites.

KEYWORDS: Fundamental electrochemistry, Pt single crystals, CO displacement, OH adsorption, potential of total zero charge



INTRODUCTION

Platinum's excellent electrocatalytic properties make it the most popular and extensively studied electrocatalyst among all elements in the periodic table. Specifically, research in electrocatalysis has primarily focused on its application in electrochemical systems for energy conversion and storage, critical for future sustainability. Most of these electrochemical reactions are surface-sensitive, necessitating the use of Pt model surfaces, such as well-defined single crystals,¹ to study the correlations between surface structure and reactivity. In aqueous electrolytes, OH adsorption on the Pt surface is a crucial step for many important electrochemical reactions.^{2–5} Numerous studies have shown that the alcohol oxidation reaction at fuel cell anodes only occurs in the presence of adsorbed OH.^{4,6,7} Similarly, OH is considered an essential intermediate for the oxygen reduction or hydrogen evolution reactions in fuel cell and battery cathodes.^{8–10}

Historically, it was initially proposed in the early studies on well-defined Pt single crystals in the 1980s that only hydrogen adsorbs on Pt at low potentials (in the so-called hydrogen adsorption region), irrespective of surface sites, both in sulfuric acid and also in the absence of specific adsorbing anions, such as in perchloric acid solutions.^{11,12} This picture was modified in the 1990s after the charge displacement experiments using CO as neutral probes where the presence of OH adsorbed on step sites at low potentials was suggested.^{13,14} However, most

macroscopic interfacial studies on Pt electrodes have disregarded possible adsorption processes other than H_{ads} in this low potential window. Moreover, recent investigations have suggested that the local charge at low-coordinated defect sites (or step sites) at low potentials disrupts the water network order.^{15–17} The coadsorption of hydrogen and oxygenated species on these step sites has been tentatively proposed as an explanation.^{18–21} The latter was experimentally confirmed in 2022. Our team demonstrated, through electrochemical and spectro-electrochemical methods that OH adsorbs on low-coordinated Pt sites with (100) geometry at low potentials.²² This study on a Pt(311) surface, which consists of two-atom-wide (111) terraces separated by monatomic (100) steps, quantified OH coverage on the (100) steps from voltammetric and charge displacement measurements analyzed using the hard-sphere model. However, whether OH also adsorbs on low-coordinated sites at low potentials on surfaces with different geometries and step densities remains unclear. In this study, not only is it confirmed that OH consistently

Received: September 27, 2024

Revised: October 25, 2024

Accepted: November 7, 2024

Published: November 18, 2024

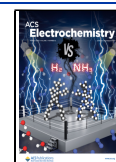
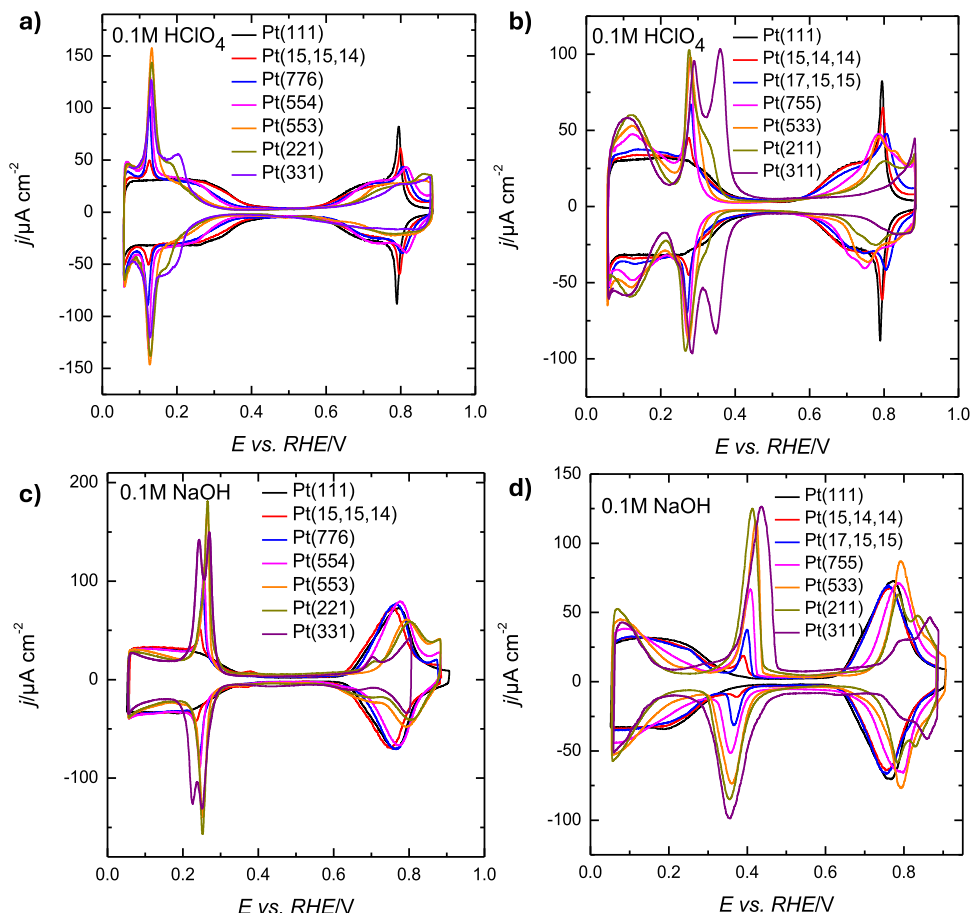


Table 1. Notation, Miller Indexes, and Step Density of the Different Surfaces Employed in the Work

LJS Notation	Miller Indices	Step Density
$\text{Pt(S)}[n(111) \times (111)] \equiv \text{Pt(S)}[(n-1)(111) \times (110)]$	$\text{Pt}(n, n, n-2)$	$\frac{2}{d\sqrt{3}\left(n - \frac{2}{3}\right)}$
$\text{Pt(S)}[n(111) \times (100)]$	$\text{Pt}(n+1, n-1, n-1)$	$\frac{2}{d\sqrt{3}\left(n - \frac{1}{3}\right)}$

**Figure 1.** Voltammetric profile of representative electrodes for the series (a, c) $\text{Pt(S)}[(n-1)(111) \times (110)]$ and (b, d) $\text{Pt(S)}[n(111) \times (100)]$ in (a, b) 0.1 M HClO_4 and (c, d) 0.1 M NaOH . Scan rate: 50 mV s^{-1} .

adsorbs on Pt step sites at low potentials, but also we provide a systematic calculation for OH coverage and analyze its dependence on surface geometry and electrolyte pH.

EXPERIMENTAL SECTION

Pt single crystal surfaces were prepared according to Clavilier's method.²³ Prior to any experiment, the single crystals were annealed in a gas-oxygen flame, cooled in a H_2/Ar atmosphere, and quenched in ultrapure water (Milli-Q, $18.2 \text{ M}\Omega \text{ cm}$) in equilibrium with the reductive atmosphere. This procedure leads to surface structures that are close to the nominal structure.^{24,25} The electrode surface was transferred to the electrochemical cell covered by a water drop of ultrapure water to avoid contamination with air impurities. The set of surfaces used in this study is listed in Table 1. For the notation of the surfaces, the so-called Lang, Joyner, and Somorjai (LJS) notation is employed.²⁶ In Table 1, the Miller indexes and equations to calculate the step density for each series of surfaces employed are also shown.

The electrochemical experiments were carried out in a commonly employed single-compartment electrochemical cell at room temperature, using a Pt counterelectrode, a reversible hydrogen electrode (RHE) as the reference electrode, and the corresponding Pt single crystals as the working electrode. As explained elsewhere,^{14,27} for the CO displacement experiments, CO is dosed inside the electrochemical cell at a given constant potential, at which CO is readily adsorbed on the surface (but not oxidized) displacing the species initially present on the Pt surface. Simultaneously, the transient current required during the forced desorption process to maintain the imposed potential value is recorded.

Voltammetric and current-transient curves were recorded with a signal generator (PAR 173), a potentiostat (eDAQ EA161), and a digital recorder (eDAQ, ED401). Solutions were prepared using NaOH (Merck, 99.99% Suprapur) and HClO_4 (Merck, Suprapur, 70%) dissolved in ultrapure water (Elga PureLab Ultra, $18.2 \text{ M}\Omega \text{ cm}$). Ar (N50, Air Liquide) was also employed to deoxygenate the solution; H_2 (N50, Air Liquide) was employed for the reference electrode, and CO

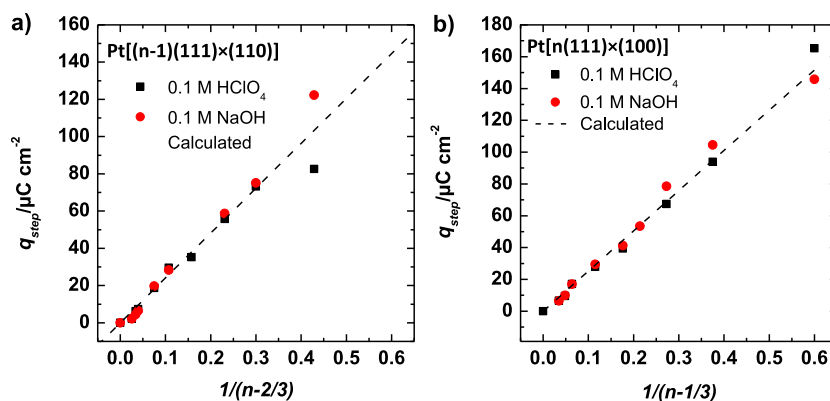


Figure 2. Experimental dependence of the charge for the step peak with the step density for (a) Pt(S)[$(n-1)(111) \times (100)$] and (b) Pt(S)[$(n-1)(111) \times (110)$] stepped surfaces in 0.1 M HClO₄ and 0.1 M NaOH. The dashed line represents the calculated values according to eqs 1 and 2.

(N50, Air Liquide) was employed for the CO displacement experiments.

RESULTS AND DISCUSSION

On the Total Coverage of the Adsorbed Species on the Steps. The adsorption energies for H_{ads} and OH_{ads} on Pt are dependent on the crystallographic surface orientation of the Pt electrode. For this reason, the voltammetric profiles of the different electrodes have distinctive features that can be used as fingerprints to characterize the distribution of sites of each geometry on the surface. Figure 1 shows the cyclic voltammetric profiles for some of the surfaces employed in this work. In the absence of specific adsorption of anions, the cyclic voltammogram for the Pt(111) surface shows two distinct adsorption processes separated by the so-called double layer. The potential region between 0.06 and 0.40 V is attributed to the H_{ads/des} process, whereas between 0.60 and 0.90 V, the OH_{ads/des} process occurs. The H_{ads/des} region on Pt(111) is nearly pH-independent, whereas the shape of the OH_{ads/des} region changes with pH. Note that the RHE shifts about 59 mV per pH unit; therefore, the invariance in the RHE scale means that the process shifts exactly the same amount as the reference electrode. The different behavior of H and OH regions has been attributed to substantial changes in the water network structure with pH, leading to different interactions between water molecules and the Pt surface.¹⁶ The structure of the water network close to the surface is also influenced by the specific interaction between the anions and their solvation layer inducing changes in the adsorption energies for OH_{ads}.²⁸

Changes can be observed in the cyclic voltammograms by introducing steps into the (111) plane. The presence of (110) steps leads to additional peaks at 0.12 and 0.24 V for pH = 1 and 13 solutions, respectively. At pH = 13, the peak slightly shifts toward positive potentials with increasing step density, evolving into two peaks for Pt(311). These peaks, whose charge increases with step density, are attributed to adsorption processes on the steps. A similar scenario is observed when (100) steps are introduced. At pH = 1, the adsorption processes on the (100) steps give rise to a reversible peak centered at 0.28 V, which increases with step density and evolves into two clear peaks at 0.28 and 0.36 V for the Pt(311) surface. In the pH = 13 solution, quasi-reversible adsorption/desorption peaks arise around 0.35 V for long terrace surfaces, with the reversibility decreasing with increasing step density. The peaks described for the steps have commonly been associated with H_{ads/des} on low-coordinated sites.¹² However,

in a recent study, we demonstrated that OH is adsorbed on (100) low-coordinated sites on a Pt(311) surface whose unit cell contains only two atoms in the terrace and one atom in the step.²² The question that now arises is whether this phenomenon is specific to the Pt(311) surface at pH = 1 or if OH is adsorbed on the steps regardless of the geometry, step density of the surface, and pH of the solution.

However, before gaining information on the nature of the adsorbed species on the step sites, a charge analysis of the peak associated with the steps is needed to determine the amount of species adsorbed on the step. Previous results in acidic solution indicate that the peak related to the steps in acidic solution involves the transfer of one electron per step site.^{29,30} As peak potential depends on pH and the nature of the cations in the supporting electrolyte,³¹ it is important to determine whether the total charge of the step processes is affected by pH. For that, experimental peak charges were obtained by integrating the charge under the step peaks in the cyclic voltammograms after subtracting the expected baseline (Figure S1) for the hydrogen adsorbed on the step and compared to theoretical values. The theoretical charge for adsorbed species on the step can be calculated by the hard-sphere model, assuming that the adsorption of one species per unit cell of the step transfers one electron.²⁹ For surfaces with n -atom wide (111) terraces and monoatomic (100) steps (Pt($n+1, n-1, n-1$)), the charge on the steps is given by the expression:³²

$$q_{100\text{step}} = \frac{2e}{\sqrt{3}d^2(n-1/3)} \quad (1)$$

where e is the charge of the electron (1.6×10^{-19} C) and d is the diameter of the Pt atom (2.77×10^{-8} cm). On the other hand, for surfaces with (111) terraces and (110) monoatomic steps (Pt($n, n, n-2$)), the charge of the step is given by³²

$$q_{110\text{step}} = \frac{2e}{\sqrt{3}d^2(n-2/3)} \quad (2)$$

From the above expressions, the theoretical representation of q_{step} vs $(n-1/3)^{-1}$ for the Pt($n+1, n-1, n-1$) series and $(n-2/3)^{-1}$ for the Pt($n, n, n-2$) series must show a slope of $\frac{2e}{\sqrt{3}d^2}$, which is equal to $241 \mu\text{C cm}^{-2}$. Figure 2 shows the theoretical and experimental dependences of the step charge with the step density for the two sets of surfaces employed at the two different pHs. To integrate the charge under the peak corresponding to the step, a reasonable base line is taken to separate it from the contribution of the terrace. As can be seen,

the experimental data in both media agrees with the theoretical values predicted by the equations, indicating that in both media an overall charge equivalent to one electron is transferred per step site. Then, assuming that the only species involved in the peaks are H_{ads} and OH_{ads} , the sum of the coverages of these species should be equal to 1.

On the Presence of Adsorbed OH on the Peak for the Step. Once the total coverage of the adsorbed species on the step is known, the nature of the species involved in the peaks can be analyzed. To answer this question, CO displacement experiments were conducted.³³ The charge density obtained from the integration of the transient current that flows during the displacement, $q_{dis}(E)$, can be understood as

$$q_{dis}(E) = q_{CO}(E) - q_{Pt}(E) \quad (3)$$

where $q_{CO}(E)$ is the electrode charge density for the CO-covered electrode at this potential (final state of the system) and $q_{Pt}(E)$ is the interfacial charge density for the Pt surface at the initial state of the experiment. Since no adsorbed species remain on the surface if the displacement is complete and the surface is completely covered by CO, $q_{CO}(E)$ is only capacitive and can be denoted as free charge density. On the other hand, the charge density required to build the interface at a given potential for platinum is equal to

$$q_{Pt}(E) = q_{free}(E) + q_H(E) + q_{OH}(E) \quad (4)$$

where $q_{free}(E)$ is the free charge density of the electrode, $q_H(E)$ is the negative charge density required to adsorb H present on the surface, which may include contributions from the terrace and the step at this potential, and $q_{OH}(E)$ is the positive charge density required to adsorb OH, present at the potential of the experiment, which may contain contributions from the step at this potential. The charge density difference between two electrode potentials for any Pt electrode can be obtained using the integration of the measured current density according to

$$q(E_2) - q(E_1) = \int_{E_1}^{E_2} \frac{j}{\nu} dE \quad (5)$$

where ν is the scan rate.

By combining eqs 3 and 4, the CO displaced charge is

$$q_{dis}(E) = q_{CO}(E) - q_{free}(E) - q_H(E) - q_{OH}(E) \quad (6)$$

If the amounts of adsorbed H and OH at a given potential are to be obtained, the values of $q_{CO}(E)$ and $q_{free}(E)$ have to be determined. However, these values are not known. Only for the Pt(111) surface, some estimations have been done. For the Pt(111) CO-covered surface, the potential of zero free charge, $E_{CO,pzfc}$, has been estimated using the known value of the work function and later measured with the immersion method.^{34,35} Thus, $q_{CO}(E)$ is then:

$$q_{CO}(E) = C_{CO}(E - E_{CO,pzfc}) \quad (7)$$

where C_{CO} is the differential capacitance of the CO surface, considered constant, which can be measured using voltammetry. Using this value and combining eqs 5 and 6, the charge for the Pt(111) electrode in the region between 0.4 and 0.6 V can be calculated, and $q(E_2)$ from the displaced charge at E_1 located in the hydrogen region. Since there is neither H nor OH adsorbed on the surface if $0.4 < E_2 < 0.6$, $q(E_2)$ can be considered the free charge in this potential window. The extrapolation of the charge in this region either considering constant values of double layer capacitance³⁶ or using the

capacitance values extracted from thermodynamic measurements³⁷ has allowed us to determine the potential of zero free charge of the Pt(111) electrode.³⁸ However, for the remaining surfaces, the values of $q_{CO}(E)$ are not known, and some reasonable approximations must be made. The most reasonable approach is to calculate $q_{CO}(E)$ by considering that the potential of zero charge varies with the introduction of steps in a similar way as the work function of the clean surface decreases with the step density. The validity of this approach has been discussed previously.³⁹ However, to extract q_H and q_{OH} from the total charge curve, $q_{free}(E)$ is still needed, and this term cannot be calculated using any of the experimental tools developed so far since it requires separating capacitive from adsorption phenomena. This value is demonstrated to be relatively small for Pt(111) at pH = 1, at least at potentials close to the potential of zero free charge, and can be considered negligible compared to $q_{dis}(E)$.⁴⁰ Figure S2 shows the total charge density curves of the different surfaces under study for pH = 1 after correcting and before $q_{dis}(E)$ by $q_{CO}(E)$. The trends are very similar for both corrected and uncorrected curves, with the only difference being that, after correction, all the curves are shifted to slightly more negative charge, but to a very similar extent, regardless of the step density for both sets of surfaces. This is reasonable considering that the $q_{CO}(E)$ values used to correct $q_{dis}(E)$ range from $-13 \mu C cm^{-2}$ for the Pt(111) surface to approximately $-11.8 \mu C cm^{-2}$ for Pt(331) and approximately $-10.1 \mu C cm^{-2}$ for Pt(311). Thus, correcting q_{dis} by q_{CO} has minimal consequence when it comes to calculating the q_{OH} dependence with step density, since the differences in the charge density curves after the implemented corrections are probably smaller than the experimental errors themselves.

For pH 13, the scenario is more complex since at this pH the $q_{free}(E)$ term cannot be considered negligible, since for the same potential in the RHE scale, the absolute potential in the SHE scale is more negative, while the potential of zero free charge is expected to remain nearly constant with pH. In other words, the distance between the potential of the displacement experiment and the potential of zero free charge increases with the pH. Hence, the value of $q_{free}(E)$ is unknown and its calculation is not straightforward. To address this issue, we compare the values of $q_{dis}(E)$ at 0.1 V for the three Pt basal planes at both pH = 1 and pH = 13 (Figure S3). It is found that this value is very similar in the two pH solutions for each of the three basal planes. If we assume that at $E = 0.1$ V only H_{ads} is displaced from the surface and that q_H is mainly governed by the geometry of the surface, and therefore it is independent of the pH, the invariance of $q_{dis}(E)$ implies that the term $q_{CO} - q_{free}$ is also pH independent or at least its variation with the pH is very small. In other words, both $q_{CO}(E)$ and $-q_{free}(E)$ increase with the increase of the pH because the potential of the experiment is more negative in an absolute reference scale; however, the increase of both terms must be similar, and it cancels out when they are subtracted. At the same time, for the Pt(100) and Pt(110) electrodes at pH = 1, it has been demonstrated that the surface is saturated with H at potentials close to the onset of hydrogen evolution.^{13,41} The displaced charges for these electrodes can then be compared to the expected values for the adsorption of one H atom on each surface site, q_{theor} . As can be seen in Table 2, $-q_{dis}(0.1 V)$ in both cases and both electrolytes are very close to the expected value, indicating that the correction $q_{CO}(E) - q_{free}(E)$ is small, and no significant differences are observed in this correction

Table 2. Comparison between the CO Displaced Charge Density for the Pt(100) and Pt(110) Electrodes in Acidic and Alkaline Solutions with the Theoretical Charge Density for a Process That Transfers One Electron per Pt Site

	$q_{\text{dis}}(0.1 \text{ V})$ in 0.1 M $\text{HClO}_4/\mu\text{C cm}^{-2}$	$q_{\text{dis}}(0.1 \text{ V})$ in 0.1 M $\text{NaOH}/\mu\text{C cm}^{-2}$	$q_{\text{theor}}/\mu\text{C cm}^{-2}$
Pt(100)	195 ± 2	200 ± 2	209
Pt(110)	148 ± 2	153 ± 2	151

regarding the nature of the electrode. The Pt(111) electrode cannot be used in this comparison, because at 0.1 V, the H adlayer has not been completed, as the large currents at this potential recorded in the voltammetric profile indicate.

Thus, since we have just concluded that the term $q_{\text{CO}}(E) - q_{\text{free}}(E)$ is nearly independent of pH and that this term is small enough at pH = 1, we also conclude that at pH = 13 this term is nearly negligible. Therefore, having established that the displaced charge density, with minor corrections, corresponds to the charge density associated with the displacement of adsorbed species on the surface, the nature of different transients can be analyzed. Figures 3 and 4 show two representative examples of the different transients measured for both series of stepped surfaces in acidic and alkaline media. As can be seen in these figures, the transients are negative at 0.1 V, indicating that H is adsorbed on the surface, whereas after the peak a positive charge is measured, indicating that OH is probably adsorbed at these potentials. Moreover, in potentials in the middle of the peak, the transient is bipolar, with an initial negative current followed by a positive current before reaching the zero current value. These bipolar transients indicate that, at these potentials, two different contributions to the displaced charge are present, and these species are displaced at different rates by CO. This is a clear indication that both OH and H are present at these potentials on the surface. The conclusion of these experiments for the (111) stepped surfaces is that, at 0.1 V, adsorbed H is present both on the terrace and step sites, and at potentials around 0.4 V in acidic and 0.45 V in alkaline solutions, OH is present on the

step sites. Above 0.6 V, OH adsorption on the (111) terrace will take place.

Determination of the OH Coverage on the Steps. For the determination of the OH coverage on the different steps, the total charge density at the interphase at 0.4 V in acidic and 0.45 V in alkaline solution should be measured. These potentials have been selected because, in the voltammetric profiles, an apparent double-layer current is obtained for all of the surfaces above these potentials. However, the displacement experiment cannot be carried out at such high potential values, because CO is readily oxidized. Even if the oxidation rate is small, it can significantly interfere with the measurement of the displaced charge. Thus, to obtain these values, the charge density differences integrated with the voltammetric profiles and the displaced charge density at 0.1 V will be used to calculate the whole curve for the displaced charge density. Thus,

$$q(E) = q(0.1 \text{ V}) + \int_{0.1 \text{ V}}^E \frac{j}{v} dE \quad (8)$$

$$q(E) = q_{\text{CO}}(0.1) - q_{\text{dis}}(0.1) + \int_{0.1 \text{ V}}^E \frac{j}{v} dE \quad (9)$$

Since, as discussed above, $q_{\text{CO}}(0.1 \text{ V})$ is not known, we defined the following approximation to $q(E)$

$$q^*(E) = -q_{\text{dis}}(0.1) + \int_{0.1 \text{ V}}^E \frac{j}{v} dE = q(E) - q_{\text{CO}}(0.1 \text{ V}) \\ = q_{\text{free}}(E) + q_{\text{H}}(E) - q_{\text{OH}}(E) - q_{\text{CO}}(0.1 \text{ V}) \quad (10)$$

where $q^*(E)$ is the charge density calculated using $q_{\text{dis}}(0.1 \text{ V})$ as the reference value for the integration, i.e., neglecting $q_{\text{CO}}(0.1 \text{ V})$.³⁸

$q^*(E)$ curves for the two series of stepped surfaces and electrolytes are shown in Figures S2 and S4. From these curves, the charge density values at 0.4 V in acidic media and 0.45 V in alkaline solutions were measured and plotted again vs $(n - 1/3)^{-1}$ for the Pt($n + 1, n - 1, n - 1$) series and $(n - 2/3)^{-1}$ for Pt($n, n, n - 2$), which are quantities that are proportional to the step density (Figure 5). As can be seen, a linear

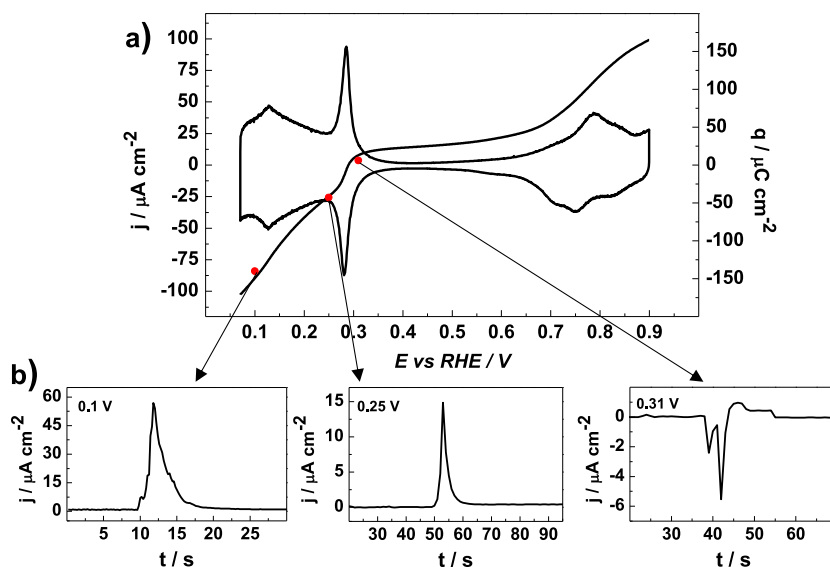


Figure 3. (a) Cyclic voltammogram and $-q_{\text{dis}}^*(E)$ curves for the Pt(755) surface as a function of the potential, recorded at a scan rate of 50 mV s^{-1} and (b) CO displacement experiments at different potentials in 0.1 M HClO_4 solution.

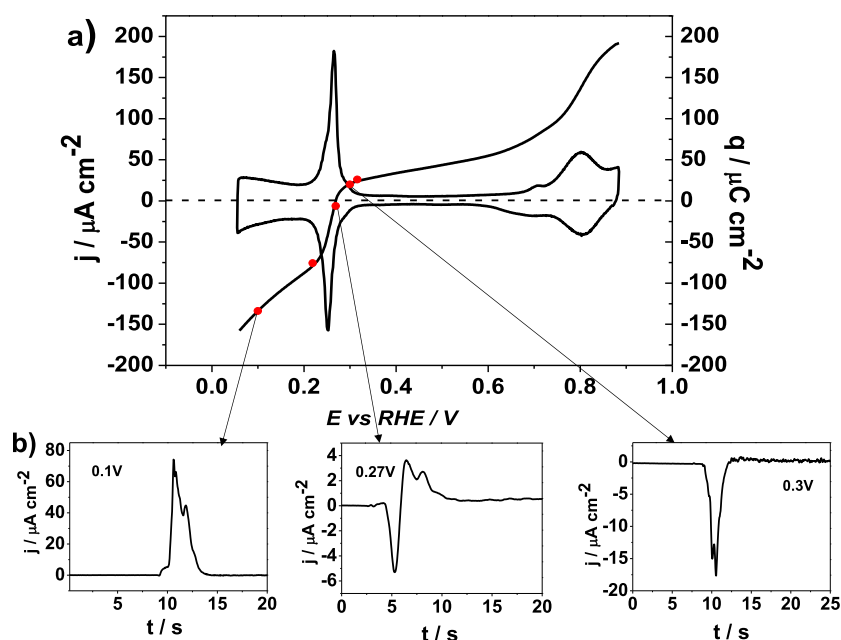


Figure 4. (a) Cyclic voltammogram and $-q_{\text{dis}}^*(E)$ curves for Pt(221) surface as a function of the potential, recorded at a scan rate of 50 mV s^{-1} and (b) CO displacement experiments at different potentials in 0.1 M NaOH solution.

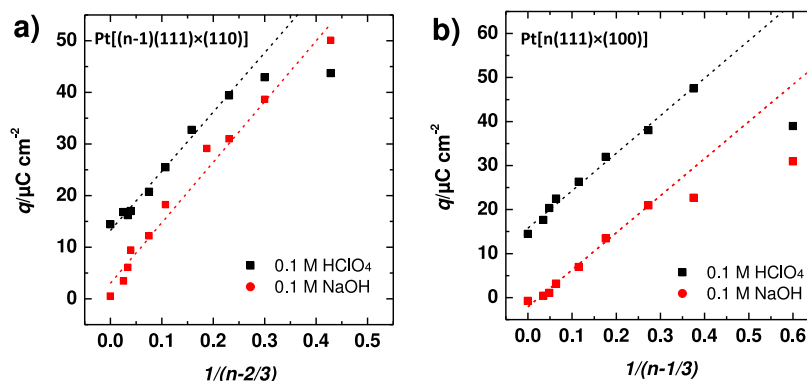


Figure 5. Charge density at 0.40 V for pH = 1 and 0.45 V for pH = 13, obtained from the charge density curves in Figure 3 and plotted versus the step density for (a) Pt(S)[$(n-1)(111) \times (110)$] and (b) Pt(S)[$n(111) \times (100)$] stepped surfaces.

relationship is obtained for both series and electrolytes. Several conclusions can be extracted from these graphs. First, the coverage of OH and H on the Pt(111) electrode at these potentials is assumed to be 0, and thus, $q^*(E)$ values for this electrode reflect the values of $q_{\text{CO}}(0.1 \text{ V}) - q_{\text{free}}(E)$, which is assumed to be almost independent of the nature of the electrode. The independence of this difference in charges with the step density is substantiated in the following equations:

$$q_{\text{CO}}(0.1 \text{ V}) - q_{\text{free}}(E) = q_{\text{CO},111}(0.1 \text{ V}) + C^{\text{CO}} \Delta E_{\text{pzfc}}^{\text{CO}} - q_{\text{free},111}(E) - C \Delta E_{\text{pzfc}} \quad (11)$$

$$q_{\text{CO}}(0.1 \text{ V}) - q_{\text{free}}(E) = q_{\text{CO},111}(0.1 \text{ V}) - q_{\text{free},111}(E) + C^{\text{CO}} \Delta E_{\text{pzfc}}^{\text{CO}} - C \Delta E_{\text{pzfc}} \quad (12)$$

If $C^{\text{CO}} \sim C$ and $\Delta E_{\text{pzfc}}^{\text{CO}} \approx \Delta E_{\text{pzfc}}$, then

$$q_{\text{CO}}(0.1) - q_{\text{free}}(E) = q_{\text{CO},111}(0.1) - q_{\text{free},111}(E) \quad (13)$$

The more negative values measured in alkaline solutions, again, are due to the change in the absolute potential, which leads to a more negative value of $q_{\text{CO}}(0.1 \text{ V}) - q_{\text{free}}(E)$. The linearity of the plots in Figure 5 validates the assumption above. Otherwise, if the difference $q_{\text{CO}}(0.1) - q_{\text{free}}(E)$ varied with the step density, a deviation of the linearity would be observed.

The increase in the value of $q^*(E)$ with the step density indicates that the amount of OH adsorbed on the electrode surface is proportional to the step density. From the slope, the OH coverage of the step sites can be calculated. As mentioned before, a slope of $241 \mu\text{C cm}^{-2}$ should be obtained for a coverage of 1. Slopes of approximately $119 \mu\text{C cm}^{-2}$ and $124 \mu\text{C cm}^{-2}$ were obtained for the Pt[$(n-1)(111) \times (110)$] series in acidic and alkaline media, respectively, for $n > 4$ surfaces. Considering the small difference of $5 \mu\text{C cm}^{-2}$ for both pH values and the inherent error in the experimental values, this difference can be considered negligible enough to establish that this slope is not dependent on the pH. By comparing this value with the theoretical slope of $241 \mu\text{C cm}^{-2}$ derived from the hard-sphere model equations, an OH

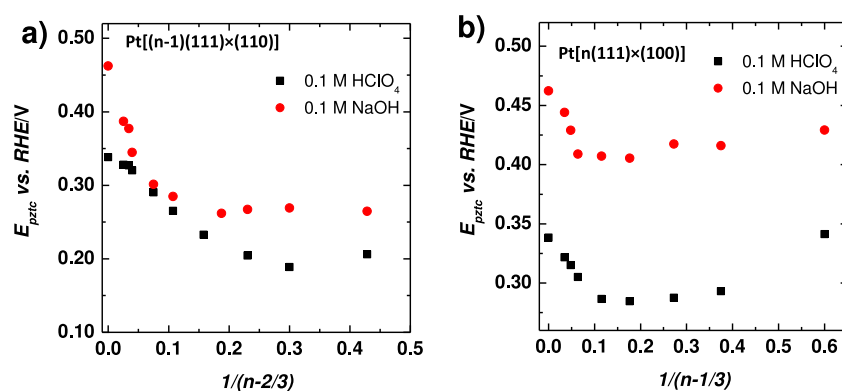


Figure 6. Potential of total zero charge, obtained from the charge density curves in Figure 3 and plotted versus the step density for (a) Pt(S)[(n − 1)(111) × (110)] and (b) Pt(S)[n(111) × (100)] stepped surfaces.

coverage of approximately 0.5 for the Pt[(n − 1)(111) × (110)] surfaces can be calculated, regardless of the pH of the electrolyte. For the Pt[n(111) × (100)] set of surfaces, a slope of 80 $\mu\text{C cm}^{-2}$ and 85 $\mu\text{C cm}^{-2}$ was obtained for acidic and alkaline media, respectively. Similar to the surfaces with (110) steps, this series of surfaces exhibits a pH-independent slope with a different θ_{OH} of approximately 0.33. Thus, it can be stated that θ_{OH} on low-coordinated sites at low potentials is pH independent but dependent on the surface structure, with higher values on (110) steps than on (100) steps. However, it is worth pointing out that, for $n < 4$ surfaces, a deviation from linearity is observed. This phenomenon was previously attributed to the interference between stress and electrical fields at closely spaced steps.⁴² However, more recent findings reveal the presence of double-height (bunched) steps and terraces with twice the nominal width for Pt surfaces when $n < 7$,⁴³ rather than uniformly spaced monoatomic steps. This observation likely accounts for the deviation from linearity in Figure 5 for $n < 4$, which may also be indicative of a different OH adsorption stoichiometry on these surfaces.⁴⁴

CO charge displacement experiments also facilitate the determination of E_{pztc} for each surface, aiming to obtain further insights into the surface charge influenced by step density and the adsorbed species at the step sites. Figure 6 illustrates the variation of E_{pztc} for Pt[(n − 1)(111) × (110)] (Figure 6a) and Pt[n(111) × (100)] (Figure 6b) sets of surfaces in both pH = 1 and pH = 13 solutions, extracted from charge density curves where $q_{\text{dis}}(E)$ is not corrected by $q_{\text{CO}}(E)$, because of the reasons discussed above.

According to this figure, the E_{pztc} values are lower for the pH 1 solution compared to the pH 13 solution for both sets of surfaces. Furthermore, E_{pztc} consistently lies at the potential for the peak of the steps (Figure 1). Given that the term $q_{\text{CO}}(E) - q_{\text{free}}(E)$ from eq 6 is shown to make a minimal contribution in the charge density curves for both pH values, it is reasonable to infer that adsorbed species on the steps predominantly influence E_{pztc} and that it is reached once the net charge is balanced on the CO-covered surface by the charge involved in OH_{ads} and H_{ads} on low-coordinated sites.

When the contribution of the terrace is relatively high ($n > 5$), the substantial effect of the terrace leads to significant variations in E_{pztc} with the step density, resulting in a rapid decrease of this value as the terrace length shortens. However, for surfaces with shorter terraces, E_{pztc} reaches a plateau. As mentioned in Figure 5, a different OH-adsorption stoichiometry is expected for these surfaces when compared to longer

terrace surfaces. Considering that E_{pztc} is dominated by the adsorbed species on the steps and that the peaks for the step in the cyclic voltammetric profiles do not shift significantly with step density (Figure 1), a plateau suggests that θ_{OH} on the steps for $n > 5$ surfaces does not differ much from θ_{OH} for longer terrace surfaces.

CONCLUSIONS

In summary, this work demonstrates that OH consistently adsorbs on Pt step sites regardless of the step geometry or pH of the electrolyte. The OH coverage on these sites was quantified and found to depend on the surface structure while being independent of pH. The potential of zero total charge was also analyzed and shown to be influenced primarily by the adsorbed species on the steps. This study provides a deeper understanding of the fundamental mechanisms governing OH adsorption on Pt surfaces that are crucial for various electrochemical applications in energy conversion and storage. These valuable insights can be ultimately essential for the design and optimization of platinum-based electrocatalysts.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acselectrochem.4c00107>.

Cyclic voltammetries with the baselines used to integrate the peak for the step and charge density curves before/after correction by q_{CO} (PDF)

AUTHOR INFORMATION

Corresponding Authors

Rubén Rizo – Instituto de Electroquímica, Universidad de Alicante, E-03080 Alicante, Spain; orcid.org/0000-0001-8161-2989; Email: ruben.rizo@ua.es

Enrique Herrero – Instituto de Electroquímica, Universidad de Alicante, E-03080 Alicante, Spain; orcid.org/0000-0002-4509-9716; Email: herrero@ua.es

Victor Climent – Instituto de Electroquímica, Universidad de Alicante, E-03080 Alicante, Spain; orcid.org/0000-0002-2033-5284; Email: victor.climent@ua.es

Authors

Lorena Chico-Mesa – Instituto de Electroquímica, Universidad de Alicante, E-03080 Alicante, Spain

Rosa M. Arán-Ais – Instituto de Electroquímica, Universidad de Alicante, E-03080 Alicante, Spain; orcid.org/0000-0001-9892-1980

Juan M. Feliu – Instituto de Electroquímica, Universidad de Alicante, E-03080 Alicante, Spain; orcid.org/0000-0003-4751-3279

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acselecchem.4c00107>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors are thankful for the financial support from Ministerio de Ciencia, Innovación y Universidades (Spain), through project PID2022-137350NB-I00. R.R. and R.M.A.-A. also acknowledge their Ramon y Cajal Fellowships from Ministerio de Ciencia, Innovación y Universidades (Spain), with references RYC2022-037376-I and RYC2022-038206-I, respectively. R.R. also acknowledges the financial support from University of Alicante through project GRE21-21A.

REFERENCES

- (1) Climent, V.; Feliu, J. M. Thirty Years of Platinum Single Crystal Electrochemistry. *J. Solid State Electrochem.* **2011**, *15* (7–8), 1297.
- (2) Pinto, L. M. C.; Quaino, P.; Arce, M. D.; Santos, E.; Schmickler, W. Electrochemical Adsorption of OH on Pt (111) in Alkaline Solutions: Combining DFT and Molecular Dynamics. *ChemPhysChem* **2014**, *15* (10), 2003–2009.
- (3) Rizo, R.; Herrero, E.; Climent, V.; Feliu, J. M. On the Nature of Adsorbed Species on Platinum Single-Crystal Electrodes. *Curr. Opin Electrochem* **2023**, *38*, No. 101240.
- (4) Mekazni, D. S.; Arán-Ais, R. M.; Ferre-Vilaplana, A.; Herrero, E. Why Methanol Electro-Oxidation on Platinum in Water Takes Place Only in the Presence of Adsorbed OH. *ACS Catal.* **2022**, *12*, 1965–1970.
- (5) García, G.; Koper, M. T. M. Carbon Monoxide Oxidation on Pt Single Crystal Electrodes: Understanding the Catalysis for Low Temperature Fuel Cells. *ChemPhysChem* **2011**, *12* (11), 2064–2072.
- (6) Rizo, R.; Ferre-Vilaplana, A.; Herrero, E.; Feliu, J. M. Ethanol Electro-Oxidation Reaction Selectivity on Platinum in Aqueous Media. *ACS Sustain. Chem. Eng.* **2023**, *11* (13), 4960–4968.
- (7) Flórez-Montaña, J.; García, G.; Guillén-Villafuerte, O.; Rodríguez, J. L.; Planes, G. A.; Pastor, E. Mechanism of Ethanol Electrooxidation on Mesoporous Pt Electrode in Acidic Medium Studied by a Novel Electrochemical Mass Spectrometry Set-Up. *Electrochim Acta* **2016**, *209*, 121–131.
- (8) Ren, H.; Zhang, Z.; Geng, Z.; Wang, Z.; Shen, F.; Liang, X.; Cai, Z.; Wang, Y.; Cheng, D.; Cao, Y. Gradient OH Desorption Facilitating Alkaline Hydrogen Evolution Over Ultrafine Quinary Nanoalloy. *Adv. Energy Mater.* **2024**, *14* (25), No. 2400777.
- (9) Liu, G.; Shih, A. J.; Deng, H.; Ojha, K.; Chen, X.; Luo, M.; McCrum, I. T.; Koper, M. T. M.; Greeley, J.; Zeng, Z. Site-Specific Reactivity of Stepped Pt Surfaces Driven by Stress Release. *Nature* **2024**, *626* (8001), 1005–1010.
- (10) Yang, Y.; Agarwal, R. G.; Hutchison, P.; Rizo, R.; Soudackov, A. V.; Lu, X.; Herrero, E.; Feliu, J. M.; Hammes-Schiffer, S.; Mayer, J. M. Inverse Kinetic Isotope Effects in the Oxygen Reduction Reaction at Platinum Single Crystals. *Nat. Chem.* **2023**, *15* (2), 271–277.
- (11) Furuya, N.; Koide, S. Hydrogen Adsorption on Platinum Single-Crystal Surfaces. *Surf. Sci.* **1989**, *220* (1), 18–28.
- (12) Clavilier, J.; El Achi, K.; Rodes, A. In Situ Characterization of the Pt (S)-[n (111)/Sx (111)] Electrode Surfaces Using Electro-sorbed Hydrogen for Probing Terrace and Step Sites. *J. Electroanal. Chem. Interfacial Electrochem* **1989**, *272* (1–2), 253–261.
- (13) Domke, K.; Herrero, E.; Rodes, A.; Feliu, J. M. Determination of the Potentials of Zero Total Charge of Pt (100) Stepped Surfaces in the [011] Zone. Effect of the Step Density and Anion Adsorption. *Journal of Electroanalytical Chemistry* **2003**, *552*, 115–128.
- (14) Climent, V.; Gómez, R.; Feliu, J. M. Effect of Increasing Amount of Steps on the Potential of Zero Total Charge of Pt (111) Electrodes. *Electrochim Acta* **1999**, *45* (4–5), 629–637.
- (15) Sarabia, F. J.; Sebastián, P.; Climent, V.; Feliu, J. M. New Insights into the Pt (Hkl)-Alkaline Solution Interphases from the Laser Induced Temperature Jump Method. *Journal of Electroanalytical Chemistry* **2020**, *872*, No. 114068.
- (16) García-Araez, N.; Climent, V.; Feliu, J. Potential-Dependent Water Orientation on Pt (111), Pt (100), and Pt (110), as Inferred from Laser-Pulsed Experiments. Electrostatic and Chemical Effects. *J. Phys. Chem. C* **2009**, *113* (21), 9290–9304.
- (17) Climent, V.; Coles, B. A.; Compton, R. G.; Feliu, J. M. Coulostatic Potential Transients Induced by Laser Heating of Platinum Stepped Electrodes: Influence of Steps on the Entropy of Double Layer Formation. *Journal of Electroanalytical Chemistry* **2004**, *561*, 157–165.
- (18) Kolb, M. J.; Calle-Vallejo, F.; Juurlink, L. B. F.; Koper, M. T. M. Density Functional Theory Study of Adsorption of H₂O, H, O, and OH on Stepped Platinum Surfaces. *J. Chem. Phys.* **2014**, *140* (13), No. 134708.
- (19) Janik, M. J.; McCrum, I. T.; Koper, M. T. M. On the Presence of Surface Bound Hydroxyl Species on Polycrystalline Pt Electrodes in the “Hydrogen Potential Region” (0–0.4 V-RHE). *J. Catal.* **2018**, *367*, 332–337.
- (20) Chen, X.; McCrum, I. T.; Schwarz, K. A.; Janik, M. J.; Koper, M. T. M. Co-adsorption of Cations as the Cause of the Apparent PH Dependence of Hydrogen Adsorption on a Stepped Platinum Single-crystal Electrode. *Angewandte Chemie* **2017**, *129* (47), 15221–15225.
- (21) McCrum, I. T.; Chen, X.; Schwarz, K. A.; Janik, M. J.; Koper, M. T. M. Effect of Step Density and Orientation on the Apparent PH Dependence of Hydrogen and Hydroxide Adsorption on Stepped Platinum Surfaces. *J. Phys. Chem. C* **2018**, *122* (29), 16756–16764.
- (22) Rizo, R.; Fernández-Vidal, J.; Hardwick, L. J.; Attard, G. A.; Vidal-Iglesias, F. J.; Climent, V.; Herrero, E.; Feliu, J. M. Investigating the Presence of Adsorbed Species on Pt Steps at Low Potentials. *Nat. Commun.* **2022**, *13* (1), 1–7.
- (23) Korzeniewski, C.; Climent, V.; Feliu, J. Electrochemistry at Platinum Single Crystal Electrodes. In *Electroanalytical Chemistry A Series of Advances*; Bard, A. J., Zoski, C. G., Eds.; CRC Press: Boca Raton, 2011; Vol. 24, pp 75–170; DOI: 10.1201/b11480.
- (24) Kibler, L. A.; Cuesta, A.; Kleinert, M.; Kolb, D. M. In-Situ STM Characterisation of the Surface Morphology of Platinum Single Crystal Electrodes as a Function of Their Preparation. *Journal of Electroanalytical Chemistry* **2000**, *484* (1), 73–82.
- (25) García-Araez, N.; Climent, V.; Herrero, E.; Feliu, J. M. On the Electrochemical Behavior of the Pt(1 0 0) Vicinal Surfaces in Bromide Solutions. *Surf. Sci.* **2004**, *560* (1–3), 269–284.
- (26) Lang, B.; Joyner, R. W.; Somorjai, G. A. Low Energy Electron Diffraction Studies of Chemisorbed Gases on Stepped Surfaces of Platinum. *Surf. Sci.* **1972**, *30* (2), 454–474.
- (27) Clavilier, J.; Albalat, R.; Gomez, R.; Orts, J. M.; Feliu, J. M.; Aldaz, A. Study of the Charge Displacement at Constant Potential during CO Adsorption on Pt (110) and Pt (111) Electrodes in Contact with a Perchloric Acid Solution. *Journal of Electroanalytical Chemistry* **1992**, *330* (1–2), 489–497.
- (28) Berná, A.; Climent, V.; Feliu, J. M. New Understanding of the Nature of OH Adsorption on Pt (1 1 1) Electrodes. *Electrochem commun* **2007**, *9* (12), 2789–2794.
- (29) Clavilier, J.; El Achi, K.; Rodes, A. In Situ Probing of Step and Terrace Sites on Pt (S)-[n (111)×(111)] Electrodes. *Chem. Phys.* **1990**, *141* (1), 1–14.
- (30) Rodes, A.; El Achi, K.; Zamakhchari, M. A.; Clavilier, J. Hydrogen Probing of Step and Terrace Sites on Pt (S)-[n (111)×(100)]. *J. Electroanal. Chem. Interfacial Electrochem* **1990**, *284* (1), 245–253.

- (31) Chen, X.; McCrum, I. T.; Schwarz, K. A.; Janik, M. J.; Koper, M. T. M. Co-adsorption of Cations as the Cause of the Apparent PH Dependence of Hydrogen Adsorption on a Stepped Platinum Single-crystal Electrode. *Angew. Chem., Int. Ed.* **2017**, *56* (47), 15025–15029.
- (32) Feliú, J. M.; Herrero, E.; Climent, V. Electrocatalytic Properties of Stepped Surfaces. In *Catalysis in Electrochemistry*; John Wiley & Sons, Inc.: Hoboken, 2011; pp 127–163.
- (33) Climent, V.; Gómez, R.; Orts, J. M.; Feliu, J. M. Potential of Zero Total Charge of Platinum Single Crystal Electrodes. In *Electrochemical Society Proceedings*; The Electrochemical Society, Inc.: Pennington, NJ, 2000; Vol. 16, p 2000.
- (34) Weaver, M. J. Potentials of Zero Charge for Platinum (111)–Aqueous Interfaces: A Combined Assessment from in-Situ and Ultrahigh-Vacuum Measurements. *Langmuir* **1998**, *14* (14), 3932–3936.
- (35) Cuesta, A. Measurement of the Surface Charge Density of CO-Saturated Pt(111) Electrodes as a Function of Potential: The Potential of Zero Charge of Pt(111). *Surf. Sci.* **2004**, *572* (1), 11–22.
- (36) Gómez, R.; Climent, V.; Feliu, J. M.; Weaver, M. J. Dependence of the Potential of Zero Charge of Stepped Platinum (111) Electrodes on the Oriented Step-Edge Density: Electrochemical Implications and Comparison with Work Function Behavior. *J. Phys. Chem. B* **2000**, *104* (3), 597–605.
- (37) Climent, V.; García-Araez, N.; Herrero, E.; Feliu, J. Potential of Zero Total Charge of Platinum Single Crystals: A Local Approach to Stepped Surfaces Vicinal to Pt(111). *Russian Journal of Electrochemistry* **2006**, *42* (11), 1145–1160.
- (38) Rizo, R.; Sitta, E.; Herrero, E.; Climent, V.; Feliu, J. M. Towards the Understanding of the Interfacial PH Scale at Pt (1 1 1) Electrodes. *Electrochim Acta* **2015**, *162*, 138–145.
- (39) Gómez, R.; Climent, V.; Feliu, J. M.; Weaver, M. J. Dependence of the Potential of Zero Charge of Stepped Platinum (111) Electrodes on the Oriented Step-Edge Density: Electrochemical Implications and Comparison with Work Function Behavior. *J. Phys. Chem. B* **2000**, *104* (3), 597–605.
- (40) Sebastián, P.; Martínez-Hincapié, R.; Climent, V.; Feliu, J. M. Study of the Pt (111) Electrolyte Interface in the Region Close to Neutral PH Solutions by the Laser Induced Temperature Jump Technique. *Electrochim Acta* **2017**, *228*, 667–676.
- (41) Ogasawara, H.; Ito, M. Hydrogen Adsorption on Pt (100), Pt (110), Pt (111) and Pt (1111) Electrode Surfaces Studied by in Situ Infrared Reflection Absorption Spectroscopy. *Chem. Phys. Lett.* **1994**, *221* (3–4), 213–218.
- (42) Ross, P. N. *The Role of Defects in the Specific Adsorption of Anions on Pt (111)*; LBNL Report #: LBL-30174; Lawrence Berkeley National Laboratory, 1990.
- (43) Valls Mascaró, F.; Koper, M. T. M.; Rost, M. J. Step Bunching Instability and Its Effects in Electrocatalysis on Platinum Surfaces. *Nat. Catal* **2024**, 1–8.
- (44) Mascaró, F. V.; Koper, M. T. M.; Rost, M. J. Quantitative Study of Electrochemical Adsorption and Oxidation on Pt (111) and Its Vicinal Surfaces. *Electrochim Acta* **2024**, *506*, No. 145014.



CAS BIOFINDER DISCOVERY PLATFORM™

STOP DIGGING THROUGH DATA —START MAKING DISCOVERIES

CAS BioFinder helps you find the
right biological insights in seconds

Start your search

CAS
A Division of the
American Chemical Society